Band-1 Hadamard Order 9344 – Classical Paley-I (q=9343); IN SPK via Paley-I; Gap 2052 Still OPEN

H_9344 Paley-I PASS; Paley-II SKIP (q=4671=3^3*173 NOT prime power); Kronecker FAIL/SKIP (4*2336; 2336===0 mod 4 admissible but H_2336 NOT in ONBOX_SEEDS); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-21

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-21_Band1_Hadamard_9324_PaleyI`; `2026-09-21_Band1_Hadamard_9320_PaleyI`; `2026-09-21_Band1_Hadamard_9312_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_9344_paleyI/`.
Cutoff. 2026-09-22T02:42:01Z (~10:42PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 9344 only (verified max|HH^T-nI|=0), via classical Paley-I q=9343 (prime ===3 mod 4). 9344 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 9344: Paley-I **PASS** ($q = 9343$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL** ($q = 4671 = 3^3 \cdot 173$ NOT prime power); Kronecker **FAIL/SKIP** (note $9344 = 4 \cdot 2336$; $2336 \equiv 0 \pmod 4$ admissible but H_{2336} NOT in ONBOX_SEEDS). **9344** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **9324+9320+9312+9284+9240+9228+9204+9200+9188+9152+9104+9128+9124+9092+2220+8884-** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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Abstract

Finite construction of order **9344**.

- H_{9344} = Paley-I ($q = 9343$) – classical SPK
- Paley-II for 9344: **SKIP** ($q = 4671 = 3^3 \cdot 173$ NOT prime power)
- Kronecker for 9344: **FAIL/SKIP** ($4 \cdot 2336$; $2336 \equiv 0 \pmod 4$ admissible but H_{2336} NOT in ONBOX_SEEDS)

- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_9344.py
```

Results. overall_pass=True; units 128/128; row blocks 73/73 (block size 128); wall 660.416s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 9344 = q + 1$ with $q = 9343$ prime $\equiv 3 \pmod 4$ -> Paley-I eligible.
- Paley-II: $q = 4671 = 3^3 \cdot 173$ NOT prime power -> SKIP.
- Kronecker: $9344 = 4 \cdot 2336$; $2336 \equiv 0 \pmod 4$ admissible but H_{2336} NOT in ONBOX_SEEDS – FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_9344 Paley-I	Paley-I q=9343	0	652.4762
H_9344 Paley-II	SKIP q=4671=3ˆ3*173 NOT prime power	–	–
H_9344 Kronecker	FAIL/SKIP (4*2336; 2336===0 mod 4 admissible but H_2336 NOT in ONBOX_SEEDS)	–	–

SPK closure note

Seeds include prior SPK through **9324** lineage (**9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; **1064+2128**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **9344**. Next constructive gap **2052** (OPEN).

Quantity	Value
9344 in SPK	**True** (Paley-I q=9343)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=9371->n=9372 Paley-I DIRECT q=9371 OR q=9391->n=9392 Paley-I DIRECT q=9391 OR q=9403->n=9404 Paley-I DIRECT q=9403 OR Track B / Object 1 meter await